Physical Significance and Governing Principles of the Shape Analytics Simulators

Course project, Mechanics of Solids

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1 Purpose and structure

The Shape Analytics app is a suite of interactive solid mechanics simulators. Its aim is to make stress analysis *move*: the user changes a load, a geometry or a material and sees the stress state, the deformation and the safety margin respond immediately. Every simulator is tied to one or more chapters of the course notes (Chapters 1 to 11). This report states, chapter by chapter, the principle used, the equations implemented, and what the quantity means physically.

Each simulator provides the same features: a live diagram, analytic charts, an A/B comparison (a saved baseline is overlaid on every chart), and exports (CSV, SVG, PNG, a design specification, and a lab report). All inputs use N, mm and MPa unless the label says otherwise.

Simulator	Chapters	Main principle used
3D stress field	2, 3, 4, 5, 6	Cauchy's theorem, equilibrium, strain–displacement, Hooke's law, half-space (Boussinesq, Cerruti) solution
Mohr's circle	1, 2, 4	Tensor transformation, principal values, invariants, hydrostatic/deviatoric split
Beam analysis	3, 8, 9	Equilibrium, pure and non-uniform bending, Euler–Bernoulli and Timoshenko theory
Shaft torsion	7	Axisymmetric equilibrium and compatibility for twist
Axial loading	6, 7	Extension of a bar, superposition, thermoelastic strain
Pressure vessel	7	Inflation of a cylinder, Lamé solution
Thermal stress	6	Isotropic thermoelasticity
Failure theories	11	Maximum principal stress, strain, shear and distortional energy criteria
Energy methods	10	Strain energy, reciprocity, Castigliano's theorem
Column buckling	9	Euler buckling, effective length

Table 1: Simulators and the chapters they implement.

2 Chapter 1: Mathematical preliminaries

Stress and strain are second-order tensors. A tensor is a physical object that exists independent of coordinates; a matrix is its representation in one basis. For a rotation tensor $\boldsymbol{Q}$ the components change as $[\boldsymbol{T}]' = \boldsymbol{Q}[\boldsymbol{T}]\boldsymbol{Q}^T$, while scalar combinations such as the trace and determinant do not.

Physical significance. This is why Mohr's circle and the invariants are meaningful. The material does not care how an analyst chose the axes, so any failure rule must depend only on quantities that survive rotation.

In the app. The Mohr's circle simulator applies $\boldsymbol{Q}$ about z to the stress matrix and reports the rotated components next to the invariants.

3 Chapter 2: Traction vector and stress tensor

3.1 Traction

Cutting a body along a plane with outward unit normal $\boldsymbol{n}$ exposes a distributed force. The traction is the force per unit area,

$$\boldsymbol{t}(\boldsymbol{x};\boldsymbol{n}) = \lim_{\Delta A \to 0} \frac{\Delta \boldsymbol{F}}{\Delta A}. \tag{1}$$

Applying linear momentum balance to an infinitesimal tetrahedron, the body force and inertia terms scale with volume (h^3) and vanish faster than the surface terms (h^2). What remains is Cauchy's theorem,

$$\boldsymbol{t}_n = \boldsymbol{\sigma}\,\boldsymbol{n}, \qquad \boldsymbol{\sigma} = \sum_i \boldsymbol{t}_i \otimes \boldsymbol{e}_i. \tag{2}$$

Physical significance. Traction on every plane through a point is fixed by just three vectors (nine numbers). Normal traction $\sigma_n = \boldsymbol{n} \cdot \boldsymbol{\sigma}\boldsymbol{n}$ pulls or pushes the planes apart; the in-plane part τ slides them. Stress is therefore local force transmission, not a property of a particular plane.

3.2 Transformation

Under rotation of axes the stress matrix transforms as $[\boldsymbol{\sigma}]' = \boldsymbol{Q}[\boldsymbol{\sigma}]\boldsymbol{Q}^T$. For plane stress rotated by θ,

$$\sigma_{x'} = \tfrac{\sigma_x+\sigma_y}{2} + \tfrac{\sigma_x-\sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta, \tag{3}$$

$$\tau_{x'y'} = -\tfrac{\sigma_x-\sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta. \tag{4}$$

In the app. In the 3D stress field the applied traction on a painted area is decomposed into normal and shear parts using exactly $\sigma_n = \boldsymbol{t}\cdot\boldsymbol{n}$ and $\boldsymbol{t}_s = \boldsymbol{t} - \sigma_n\boldsymbol{n}$. The Mohr's circle simulator implements the transformation above.

4 Chapter 3: Stress equilibrium equations

Linear momentum balance for a small element gives

$$\nabla \cdot \boldsymbol{\sigma} + \rho\boldsymbol{b} = \rho\ddot{\boldsymbol{u}}, \tag{5}$$

and angular momentum balance, in the absence of body couples, gives $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$ (so $\tau_{xy} = \tau_{yx}$ and so on). The traction boundary condition is $\boldsymbol{\sigma}\boldsymbol{n} = \bar{\boldsymbol{t}}$ on the loaded surface.

Physical significance. Equilibrium says that stress cannot be chosen freely: its gradient must balance the body force, which is why stress varies smoothly through a loaded body. Symmetry of $\boldsymbol{\sigma}$ means a pair of shear stresses always acts on perpendicular faces, so six independent components suffice.

In the app. The beam simulator uses the integrated form of equilibrium, $dV/dx = -w$ and $dM/dx = V$, to draw the shear-force and bending-moment diagrams. The mohr simulator accepts only symmetric shears for the same reason.

5 Chapter 4: Principal values, maximum shear and Mohr's circle

Principal planes are the planes on which traction is purely normal, so $\boldsymbol{\sigma n} = \lambda \boldsymbol{n}$. They solve the eigenvalue problem

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0, \tag{6}$$

with the invariants

$$I_1 = \operatorname{tr}\boldsymbol{\sigma}, \quad I_2 = \sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{zx}^2, \quad I_3 = \det\boldsymbol{\sigma}. \tag{7}$$

The roots are real, the principal directions are orthogonal, and the principal stresses are the extreme values of the normal traction. The maximum shear stress is $\tau_{\max} = (\sigma_1 - \sigma_3)/2$ on planes at 45° to the extreme principal directions. In the (σ, τ) plane the state of every plane lies on or between three circles, with diameters $\sigma_1 - \sigma_3$, $\sigma_1 - \sigma_2$ and $\sigma_2 - \sigma_3$ (Mohr's circle and Mohr's stress plane). The stress splits into a hydrostatic part $p\boldsymbol{I}$ with $p = I_1/3$ (volume change) and a deviatoric part $\boldsymbol{s} = \boldsymbol{\sigma} - p\boldsymbol{I}$ (shape change).

Physical significance. Principal stresses are the largest and smallest tensile or compressive intensities a point feels, and they decide where a brittle material cracks. The maximum shear controls ductile slip. The hydrostatic part changes volume but not shape, and the deviatoric part changes shape; this separation reappears in the von Mises criterion.

In the app. The simulator solves the cubic with the trigonometric (Lode) form, draws the three circles, marks the rotating face as a point that moves twice as fast as the element, and reports I_1, I_2, I_3, p and the failure margin.

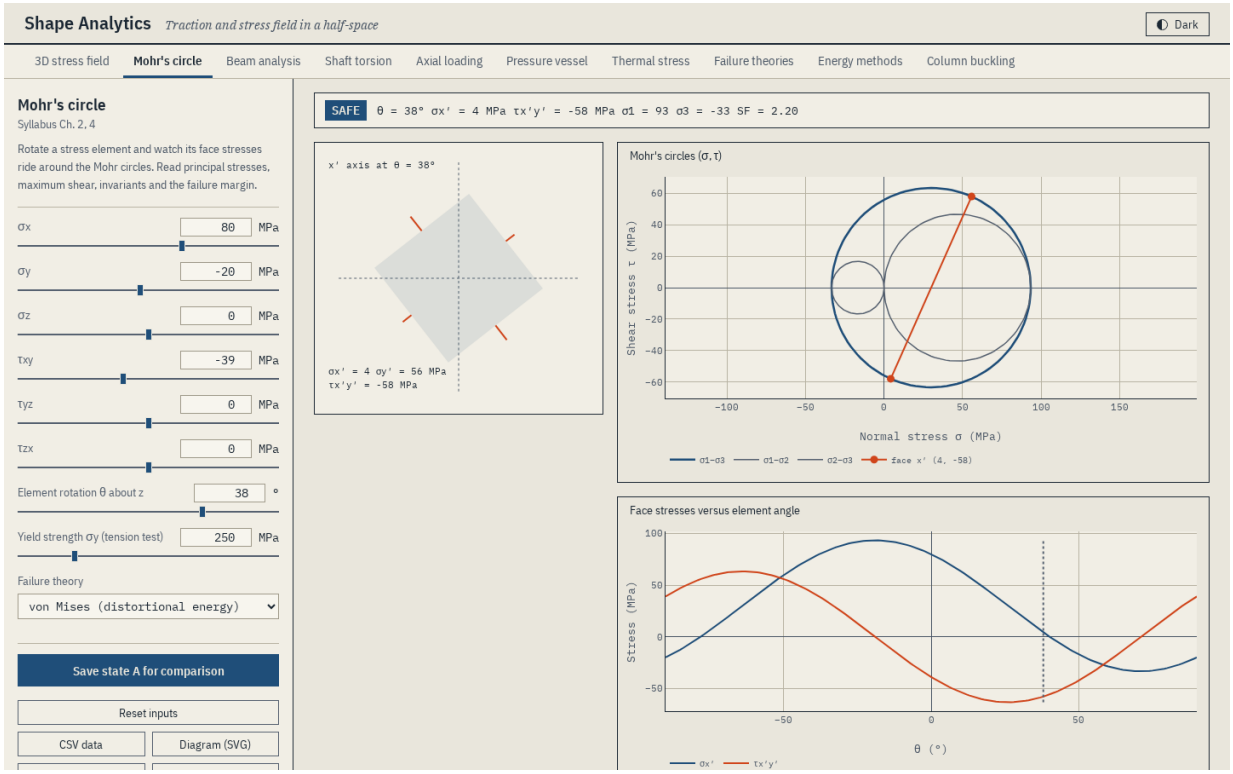


Figure 1: Mohr's circle simulator: the rotated element, the three circles and the face stresses versus angle.

6 Chapter 5: Strain tensor

Deformation is described by mapping a reference configuration to a deformed one. For small deformation the strain tensor is the symmetric part of the displacement gradient,

$$\boldsymbol{\varepsilon} = \tfrac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T\right), \tag{8}$$

with longitudinal strain $\varepsilon_{xx} = \partial u_x/\partial x$, shear strain $\gamma_{xy} = 2\varepsilon_{xy}$, and volumetric strain $\varepsilon_v = \operatorname{tr}\boldsymbol{\varepsilon}$. The antisymmetric part is a local rigid rotation. Six strain components derive from three displacements, so they must satisfy compatibility conditions or the body would tear or overlap. Strain transforms exactly like stress, so there are principal strains and a Mohr's circle for strain.

Physical significance. Strain measures how much neighbouring points move relative to each other, after removing rigid motion. It is the quantity the material actually responds to, which is why constitutive laws link stress and strain.

In the app. The 3D stress field colours the body by strain magnitude and exaggerates the displacement field computed from the half-space solution so the deformed shape is visible. Axial, torsion and thermal simulators compute normal, shear and thermal strain respectively.

7 Chapter 6: Stress–strain relations

A linear elastic solid obeys $\sigma_{ij} = C_{ijkl}\varepsilon_{kl}$. Symmetry of stress, strain and of the strain energy reduces the 81 constants to 21 for the most general material and to two for an isotropic one,

$$\sigma_{ij} = \lambda\,\varepsilon_{kk}\delta_{ij} + 2\mu\,\varepsilon_{ij}, \qquad \varepsilon_{ij} = \tfrac{1+\nu}{E}\sigma_{ij} - \tfrac{\nu}{E}\sigma_{kk}\delta_{ij}. \tag{9}$$

The engineering constants are related by $G = \mu = E/2(1+\nu)$ and $K = E/3(1-2\nu)$. With a temperature change the free strain $\alpha\Delta T\,\delta_{ij}$ is added, giving isotropic thermoelasticity $\varepsilon_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\sigma_{kk}\delta_{ij} + \alpha\Delta T\delta_{ij}$.

Physical significance. E is stiffness in tension, G is stiffness in shear, ν is the sideways contraction that accompanies stretching, and K is resistance to volume change. Heating produces strain without stress; stress appears only if the supports prevent the strain.

In the app. The material selector fixes E, ν, α and σ_y for every simulator and the readouts show G and K. The thermal stress simulator implements the constrained cases

$$\sigma_{\text{bar}} = -E\alpha\Delta T, \quad \sigma_{\text{plate}} = \frac{-E\alpha\Delta T}{1-\nu}, \quad \sigma_{\text{block}} = \frac{-E\alpha\Delta T}{1-2\nu}. \tag{10}$$

The more directions are blocked, the stiffer the effective response and the larger the stress.

8 Chapter 7: Axisymmetric problems

In cylindrical coordinates (r, θ, z) the equilibrium equation in the radial direction reads $\frac{d\sigma_{rr}}{dr} + \frac{\sigma_{rr}-\sigma_{\theta\theta}}{r} = 0$ when the fields depend on r only. The chapter solves three canonical problems for a long cylinder: extension, torsion and inflation.

8.1 Extension and torsion

For extension, $\sigma_{zz} = N/A$ and $\delta = NL/EA$. For torsion, physical symmetry says plane sections stay plane and rotate rigidly, so $\gamma_{\theta z} = r\,d\phi/dz$ and, from Hooke's law,

$$\tau(r) = \frac{Tr}{J}, \qquad J = \frac{\pi}{32}(D^4 - d^4), \qquad \phi = \frac{TL}{GJ}. \tag{11}$$

Physical significance. Torque is carried only by shear stress that grows linearly with radius, so material near the axis is under-used and a hollow shaft keeps most of the stiffness at much lower mass. Torsional stiffness GJ/L plays the same role for twist that EA/L plays for stretch.

In the app. The axial simulator handles stepped bars in series, a fixed–fixed redundant case by compatibility $\sum \delta_i = 0$, and thermal strain. The torsion simulator plots $\tau(r)$ and $\phi(x)$ and compares the stiffness of the hollow and solid sections.

8.2 Inflation of a cylinder (Lamé)

Equilibrium plus Hooke's law plus compatibility give $\sigma_{rr} = A - B/r^2$ and $\sigma_{\theta\theta} = A + B/r^2$. The boundary conditions $\sigma_{rr}(a) = -p_i$ and $\sigma_{rr}(b) = -p_o$ fix

$$A = \frac{p_i a^2 - p_o b^2}{b^2 - a^2}, \qquad B = \frac{(p_i - p_o)a^2 b^2}{b^2 - a^2}, \tag{12}$$

and, for closed ends, $\sigma_{zz} = A$. For thin walls this reduces to $\sigma_\theta = pr/t$ and $\sigma_z = pr/2t$; for a sphere $\sigma_\theta = \sigma_\phi = pr/2t$.

Physical significance. Internal pressure tries to enlarge the vessel, and the wall resists with hoop tension, exactly as a hoop around a barrel does. The hoop stress is twice the axial stress, so a pressurised cylinder splits along its length. In a thick wall the hoop stress is largest at the inner surface, where the radial stress also equals $-p_i$.

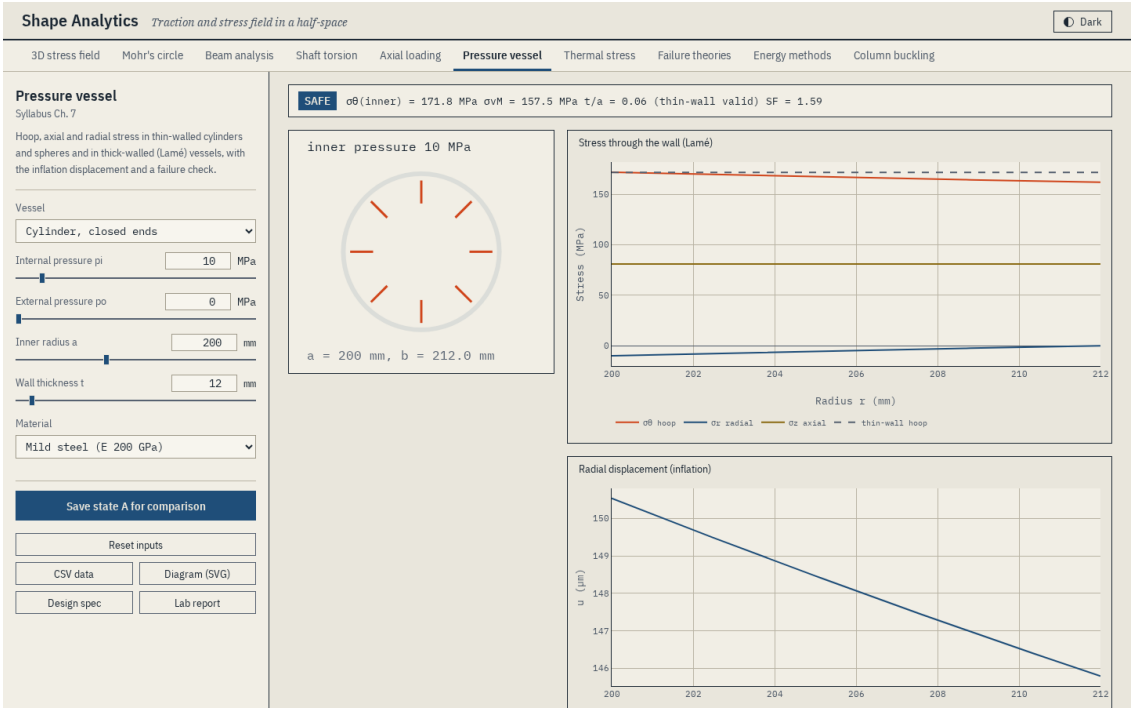


Figure 2: Pressure vessel simulator: Lamé stresses through the wall with the thin-wall hoop stress for comparison.

9 Chapter 8: Bending of beams

Pure bending. Plane sections remain plane and normal to the axis, so axial strain varies linearly with distance y from the neutral axis. With zero net axial force the neutral axis passes through

the centroid, and

$$\sigma_{xx} = -\frac{My}{I}, \qquad \frac{1}{\rho} = \frac{M}{EI}. \tag{13}$$

Non-uniform bending. When the moment varies, the shear force $V = dM/dx$ produces shear stress

$$\tau = \frac{VQ}{Ib}, \qquad Q = \int_y^c y'\, b(y')\, dy', \tag{14}$$

which is largest at the neutral axis (for a rectangle $1.5V/A$, for a solid circle $4V/3A$). Unsymmetrical sections need the product of inertia and a shear centre, the point through which the load must act to avoid twist.

Physical significance. The beam carries load by stretching one side and shortening the other, so fibres far from the neutral axis do the work, which is why deep or I-shaped sections are efficient. Shear is carried by the web, and the shear stress is zero at the outer surface.

In the app. The beam simulator solves statics for simply supported and cantilever beams under a point load, a uniform load and an applied moment, then integrates the curvature numerically. It plots V, M, deflection, and the through-depth distributions of σ and τ for rectangular, solid and hollow circular sections.

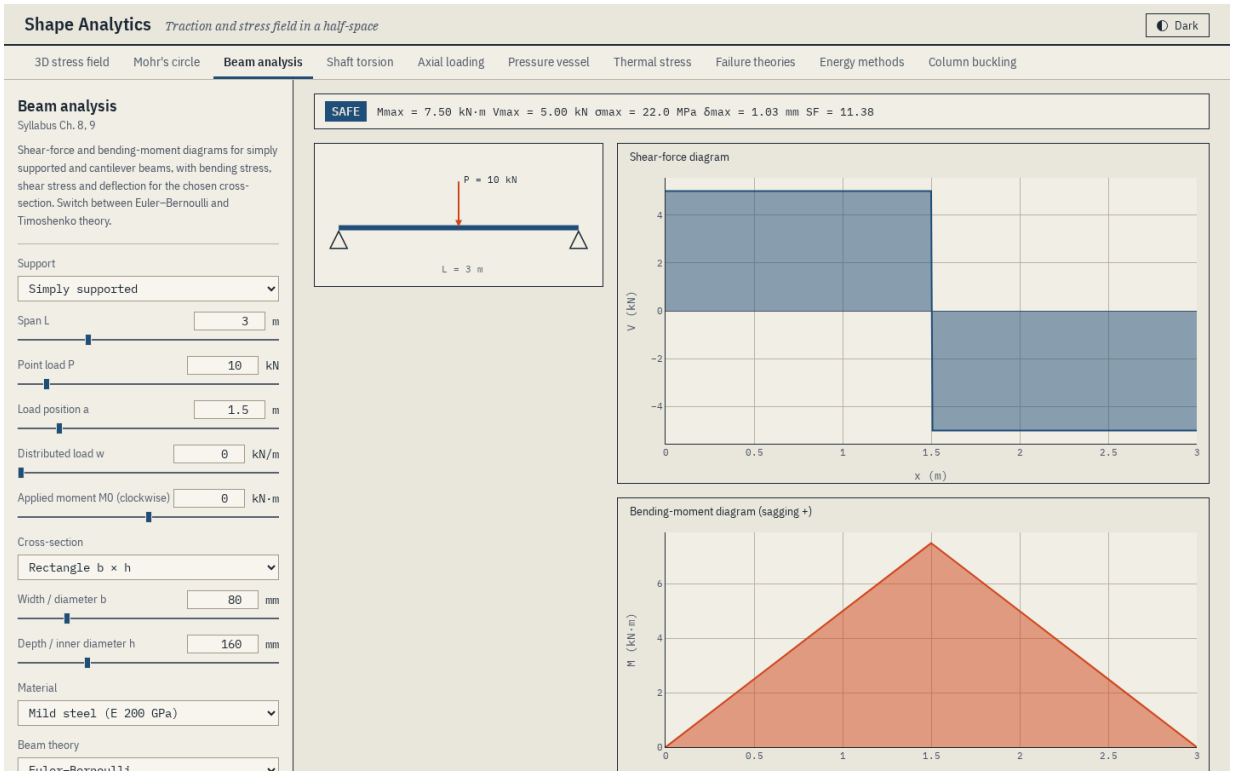


Figure 3: Beam simulator: loading diagram, shear force and bending moment.

10 Chapter 9: Theory of beams

Euler–Bernoulli theory assumes plane sections remain plane and normal to the deformed axis, giving $EI\, v'' = M(x)$. **Timoshenko theory** lets the section rotate independently of the axis, so shear deformation is retained. The extra deflection under shear is $\int V/(k_s GA)\, dx$, with shear correction factor k_s (for example $10(1+\nu)/(12+11\nu)$ for a rectangle). For a cantilever with a tip

load,

$$y_{\mathrm{TBT}} = \frac{PL^3}{3EI} + \frac{PL}{k_s GA}, \qquad \frac{y_{\mathrm{TBT}} - y_{\mathrm{EBT}}}{y_{\mathrm{TBT}}} = \frac{1}{1 + k_s GAL^2/3EI}, \tag{15}$$

so the error of Euler–Bernoulli theory grows as the beam becomes short and deep.

Buckling. A slender column under axial compression becomes unstable when

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}, \qquad \sigma_{cr} = \frac{\pi^2 E}{\lambda^2}, \qquad \lambda = \frac{KL}{r}, \tag{16}$$

with $K = 1,\ 0.7,\ 0.5,\ 2$ for pinned–pinned, fixed–pinned, fixed–fixed and fixed–free ends. Euler's formula assumes the column stays elastic. For $\lambda < \lambda_c = \pi\sqrt{E/\sigma_y}$ it would give a stress above yield, so short columns follow the Johnson parabola

$$\sigma_{cr} = \sigma_y \left[1 - \frac{\sigma_y \lambda^2}{4\pi^2 E}\right], \tag{17}$$

which meets Euler's curve at λ_c where $\sigma_{cr} = \sigma_y/2$.

Physical significance. Buckling is a loss of stability, not of strength: the straight shape stops being the stable equilibrium at a load set by bending stiffness EI and geometry, often far below the yield load. Restraining the ends shortens the effective length and raises P_{cr} with the square of $1/K$.

In the app. The beam simulator has an Euler–Bernoulli / Timoshenko switch. The column buckling simulator plots σ_{cr} against slenderness with the Euler curve, the Johnson parabola and the yield line, bases its safety factor on whichever governs, and draws the first buckled mode shape for each end condition.

11 Chapter 10: Energy methods

Work done on a linear elastic body is stored as strain energy $U = \frac{1}{2}\int \boldsymbol{\sigma} : \boldsymbol{\varepsilon}\, dV = \frac{1}{2}\sum P_i \delta_i$. For common members

$$U_{\mathrm{axial}} = \frac{N^2 L}{2EA}, \quad U_{\mathrm{torsion}} = \frac{T^2 L}{2GJ}, \quad U_{\mathrm{bending}} = \int \frac{M^2}{2EI} dx, \quad U_{\mathrm{shear}} = \int \frac{V^2}{2k_s GA} dx. \tag{18}$$

Maxwell–Betti reciprocity: $P_a \delta_{ab} = P_b \delta_{ba}$. **Castigliano's first theorem:** the displacement at a load, in the direction of the load, is

$$\delta_i = \frac{\partial U}{\partial P_i}. \tag{19}$$

Physical significance. Because stored energy depends only on the final loads and not on how they were applied, one scalar function U contains all the load–displacement information. Reciprocity is a consequence of linearity: influence is symmetric.

In the app. The energy simulator computes U for four members, differentiates $U(P)$ numerically to obtain δ, and compares it with the closed form. It also shows reciprocity on a simply supported beam, where the deflection at $0.7L$ from a load at $0.3L$ equals the reverse. Switching shear energy on reproduces the Timoshenko result of Chapter 9.

12 Chapter 11: Theory of failure

A failure theory turns a stress state into one equivalent value that is compared with the critical value σ_{cr} of a simple tension test. The criteria implemented are:

Theory	Failure condition	Best for
Maximum principal stress	$\max \lvert\sigma_i\rvert \geq \sigma_{cr}$	brittle fracture
Maximum normal strain	$\varepsilon_1 = [\sigma_1 - \nu(\sigma_2 + \sigma_3)]/E \geq \sigma_{cr}/E$	brittle materials
Maximum shear (Tresca)	$\sigma_1 - \sigma_3 \geq \sigma_{cr}$	ductile metals (conservative)
Distortional energy (von Mises)	$\sqrt{\frac{1}{2}[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2]} \geq \sigma_{cr}$	ductile metals

The octahedral shear stress is $\tau_{oct} = \frac{\sqrt{2}}{3}\sigma_{vM}$, so the octahedral shear and distortional energy theories predict the same envelope.

Physical significance. Ductile metals yield by shear, which depends only on the deviatoric part of stress: a hydrostatic pressure, however large, does not yield them. This is why Tresca and von Mises depend only on differences of principal stresses. Plotted in the σ_1–σ_2 plane, the von Mises criterion is an ellipse, Tresca a hexagon inscribed in it, and the maximum principal stress theory a square.

In the app. Every simulator that produces a stress state also reports a safety factor σ_y/σ_{eq}. The failure simulator draws all envelopes together with the operating point and reports the governing theory.

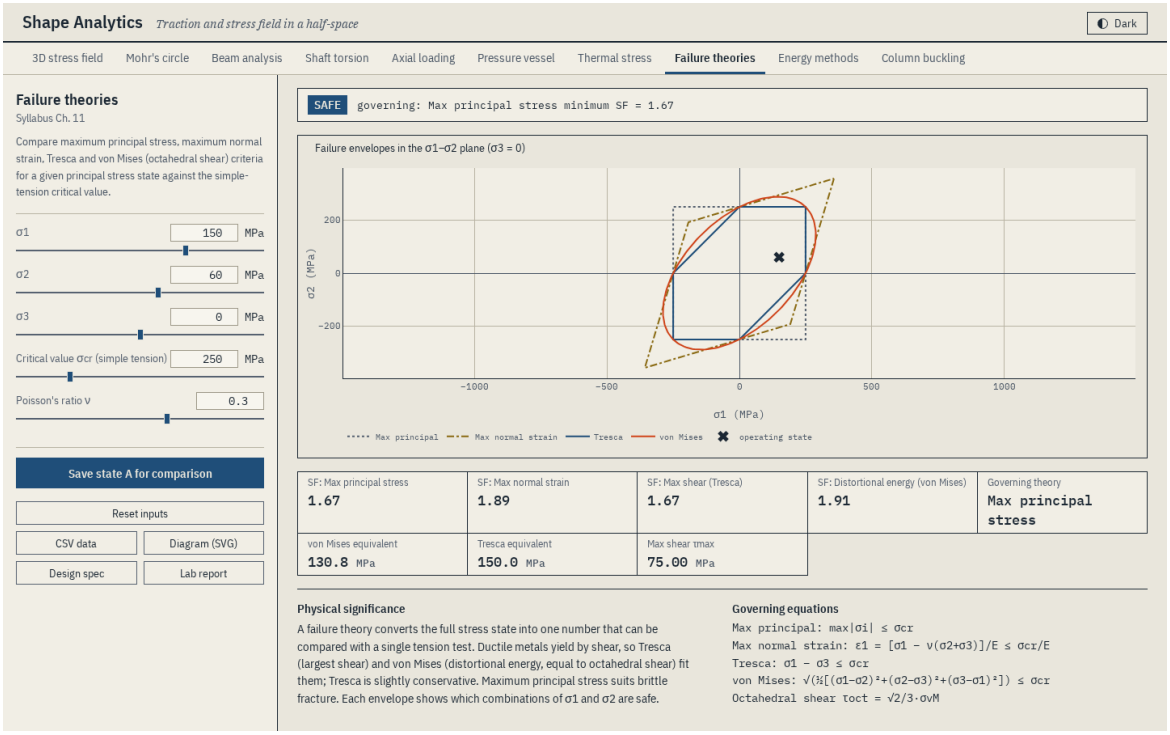


Figure 4: Failure theory simulator: envelopes in the σ_1–σ_2 plane and safety factors.

13 The 3D stress field simulator

The original module of the app combines Chapters 2 to 6 for a body loaded by a traction on an arbitrary painted area. Tension is positive and each load has a local frame with z pointing into the body ($R^2 = r^2 + z^2$).

Normal load (Boussinesq). A point force P into a half-space gives the full stress state

$$\sigma_z = -\frac{3Pz^3}{2\pi R^5}, \qquad \tau_{rz} = -\frac{3Prz^2}{2\pi R^5}, \tag{20}$$

$$\sigma_r = \frac{P}{2\pi}\left[\frac{1-2\nu}{R(R+z)} - \frac{3r^2 z}{R^5}\right], \qquad \sigma_\theta = \frac{P(1-2\nu)}{2\pi}\left[\frac{z}{R^3} - \frac{1}{R(R+z)}\right]. \tag{21}$$

The hoop stress σ_θ is not zero and enters the von Mises stress and the strain. These stresses satisfy $\nabla\cdot\boldsymbol{\sigma} = \mathbf{0}$ (Chapter 3), vanish on the free surface away from the load, and carry the load: $\int \sigma_z\, dA = -P$ on every horizontal plane.

Tangential load (Cerruti). The displacement of a half-space under a tangential point force is known in closed form. The stress follows as in Chapters 5 and 6: the strain is the symmetric gradient of the displacement, $\boldsymbol{\epsilon} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$, and Hooke's law gives $\boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\epsilon})\,\mathbf{I} + 2G\boldsymbol{\epsilon}$ with $\lambda = E\nu/[(1+\nu)(1-2\nu)]$. Since $\lambda \to \infty$ as $\nu \to 0.5$, the program caps ν at 0.499 in this formula.

Distributed loads. The painted area is a set of point loads, each with an equivalent-depth offset $z_{\text{eff}} = z + a$ that removes the singularity under a point. Because the equations are linear (Chapter 10), the *tensors* of all point loads and of separately saved loads are rotated to global axes and added; the principal stresses, von Mises stress and strain are then computed from the summed tensor, never from the maximum of scalars.

Displayed measures. $\sigma_n = \max(|\sigma_1|, |\sigma_3|)$ and $\tau_{\max} = (\sigma_1 - \sigma_3)/2$ from the principal stresses (closed-form roots of the characteristic cubic), the von Mises stress, and the strain norm from $\boldsymbol{\epsilon} = [(1+\nu)\boldsymbol{\sigma} - \nu\,\mathrm{tr}(\boldsymbol{\sigma})\mathbf{I}]/E$. Poisson's ratio follows the material (steel 0.30, aluminium 0.33, concrete 0.20, polymer 0.38, rubber 0.49). The deformed shape uses an exaggeration factor independent of E, so stiffness ratios are kept. A strain above 1% is flagged as beyond small-strain theory.

Full-surface pressure. Chapter 2 shows that a pressure p on the entire boundary gives the hydrostatic state $\boldsymbol{\sigma} = -p\,\mathbf{I}$. The app uses this only when a normal pressure covers at least 99% of the surface *area*; below that, the field is the half-space solution.

Section analytics. The drawer plots a cut through the body on a chosen plane. The server evaluates the same tensor on an 80×80 grid and returns the four maps of the 3D view. Points outside the body (box, cylinder with its axis along y, or sphere) are blank, and the slice position is limited to the extent of the body.

Physical significance. The model shows how a local load spreads with depth: stress decays rapidly away from the loaded area (St. Venant's principle) while the volume of loaded material grows. It also demonstrates that a purely normal surface traction produces shear stress below the surface, which is where yielding can start.

14 Finite element method (beyond the syllabus)

The finite element method solves the same equations numerically. The body is cut into elements joined at nodes, the displacement inside an element is interpolated from its nodal values, $\mathbf{u} = \sum_i N_i \mathbf{u}_i$, and the strain is $\boldsymbol{\epsilon} = \mathbf{B}\mathbf{u}_e$ (Chapter 5). Hooke's law gives $\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\epsilon}$ (Chapter 6), and the

principle of virtual work (Chapter 10) gives the element stiffness

$$\mathbf{k}_e = \int_{V_e} \mathbf{B}^T \mathbf{D}\, \mathbf{B}\, dV. \tag{22}$$

Adding all $\mathbf{k}_e$ gives $\mathbf{K}\mathbf{u} = \mathbf{f}$; $\mathbf{K}$ is symmetric (Maxwell–Betti reciprocity) and singular until the supports remove rigid-body motion. The traction-free condition on a free face is satisfied automatically.

Problem	Elements	Reference
Tapered bar	2-node bar	exact integral of P/EA
Cantilever beam	Hermite beam	$PL^3/3EI$ and $wL^4/8EI$
Plane-stress cantilever	CST, Q4	Euler–Bernoulli and Timoshenko
Plate with a hole	CST, Q4	Kirsch, $K_t = 3$
3D block, patch load	8-node brick	half-space (Boussinesq, Newmark)

Physical significance. Finite elements are not limited to the simple shapes of the closed-form chapters. The 3D block has real free faces and a base, so unlike the half-space solution it leaves no stress on its sides: the side-face σ_x is about 0.4% of the pressure in the FEM against 2.6% for the half-space at the default block. As the block grows the centre stress approaches the half-space value (ratio 1.26, 1.03 for a 2×3 m and an 8×8 m block), which also validates the 3D tab for deep and wide bodies.

In the app. A separate tab holds the five problems. Each shows the deformed mesh coloured by stress, the comparison with the reference, a mesh-refinement chart, and the stiffness matrix for the bar and beam. A short primer on the method is in `docs/fem_primer.pdf`.

15 Limitations

- All models are linear elastic and small-strain. Plasticity (Chapter 6.8), large rotations and dynamics are not modelled. Rubber in the 3D tab can exceed 100% strain, where the numbers are only indicative (the readout says so).
- Beam deflections are obtained by numerical integration on 401 points; the Timoshenko correction uses a constant shear factor.
- The Lamé solution assumes a long vessel with closed ends and uniform temperature.
- The half-space solution assumes an infinite body. For a finite body it leaves stress on the sides and base; the FEM block quantifies this. A partial patch load in the 3D tab has no supports, so displacements are referenced to the bottom centre.
- The column simulator assumes an ideal straight, concentrically loaded column; imperfections lower the real buckling load.
- FEM: plane stress only; CST and Q4 elements are stiff in bending until the mesh is refined (a coarse slender Q4 cantilever is about 11% too stiff, CST about 46%); 8-node bricks are stiff for nearly incompressible materials; surface stresses are extrapolated and the least accurate.

16 Verification by limiting cases

Every component was tested against closed forms and limiting cases: hydrostatic, uniaxial and pure-shear states, zero load, $\nu \to 0.5$, thin and stubby members, full coverage, and extreme inputs. The results are summarised below; the test scripts are not kept in the project.

Component	Checks	What was verified
`physics.js`	54	invariants of random tensors over 12 decades of stress; $\nabla\cdot\boldsymbol{\sigma}=\mathbf{0}$ for Boussinesq and Cerruti; stress equals Hooke's law applied to the displacement gradient; $\int\sigma_z\,dA=-P$; free surface; far field; singular points
`server.py`	40	eigen-decomposition against numpy; Cauchy relation; linearity, superposition and symmetry of the section; masks; agreement of the browser and server physics to 2×10^{-4} for four load orientations
Simulators	110	beam (simply supported, cantilever, UDL, end moment, Timoshenko limits), shaft, stepped and fixed bars, Lamé cylinder and sphere, thermal restraint, failure envelopes, Castigliano and reciprocity, Euler and Johnson
Simulators, random	22 500 inputs	no exception and no NaN; the only infinite value is a safety factor with no load
FEM	43	patch test, closed forms, equilibrium, uniform-pressure block, convergence toward the half-space, 116 extreme inputs
3D tab	94 cases	every shape, material, load type and map with painted loads, no non-finite output

Defects found and corrected.

1. *Torsion in the energy tab:* the closed-form displacement and work were off by a factor of 10^6 (a unit slip) and the status check skipped torsion. Corrected: Castigliano's derivative now matches ϕ for the shaft.

2. *Short columns:* Euler's stress reached 4×10^4 MPa and the safety factor used it. The Johnson parabola now governs for $\lambda<\lambda_c$.

3. *Principal stresses:* a fixed threshold treated tiny non-hydrostatic stress states as hydrostatic. The test is now relative to the stress scale.

4. $\nu=0.5$*:* the Lamé constant divided by zero (infinite stress in the browser, a division-by-zero error on the server). ν is capped at 0.499 there.

5. *Section endpoint:* a resolution below 2 crashed (now clamped to 2–400) and a non-unit force direction scaled the load (now normalised). Earlier in the project the section used the wrong cylinder axis and showed stress outside the body; both were corrected.

6. *Display:* an infinite safety factor appeared as a dash and now shows as ∞; strains above 1% are flagged.

Hand checks of the default cases remain consistent: a simply supported beam with a central load gives $M_{\max}=PL/4=7.5$ kN m and $\delta=PL^3/48EI=1.03$ mm; the circular shaft gives $\tau_{\max}=TR/J=32.6$ MPa and $\phi=1.17°$; the plane-stress Mohr circle gives $\sigma_{1,3}=93.4$, -33.4 MPa; the Lamé hoop stress at the inner wall agrees with pr/t to 0.1% for $t/a=0.06$; and Castigliano's derivative of U reproduces $PL^3/3EI$ for the cantilever.